# PULSE: Data, Database & Logic Guide

What exists, how it connects, who fills each field, what rings and bracelets send, how the fields match Algerian blood-pressure factors, and what we still need.

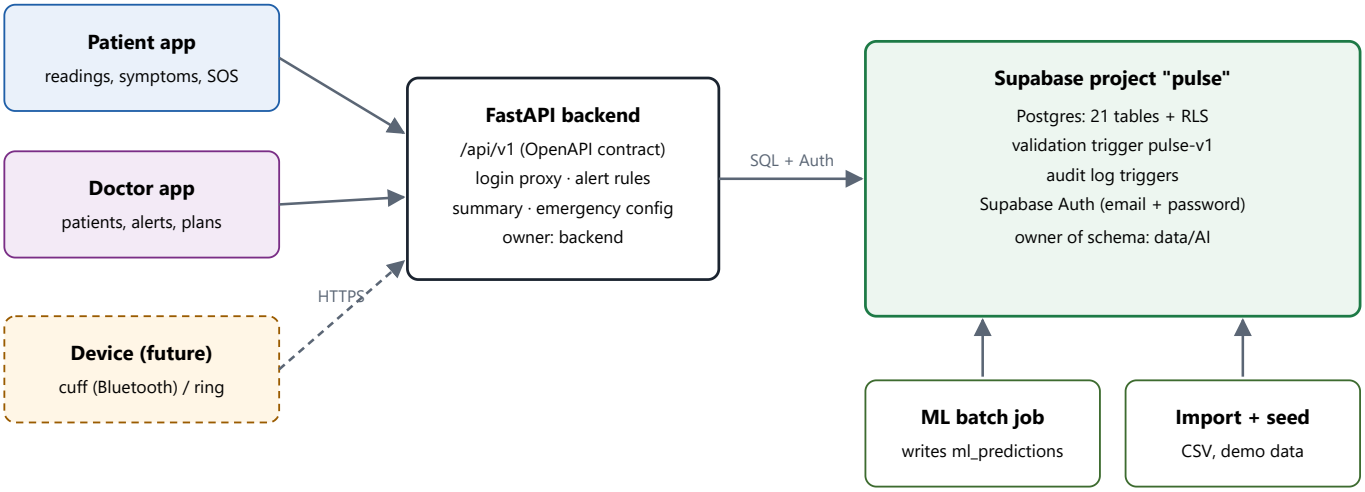
30 October 2026 · Data/AI workstream · Prototype, not a medical device. All data so far is fictional and labelled `simulated`.

**Status labels:** LIVE running in the cloud database · BUILT code exists · PROPOSAL not built, needs a team decision · MISSING nobody has it yet · TO VERIFY must be checked before use

## 1. Where we are (at a glance)

| Part | Status | Details |
|---|---|---|
| Cloud database `pulse` (Supabase, Paris) | LIVE | 21 tables, access rules (RLS) on every table, 46 policies, 6 migrations applied. Logged-out requests and public sign-up are refused. |
| Data code on GitHub, branch `AI` | BUILT | `supabase/` migrations, `pulse_data/` CSV import and demo data generator, `ml/` model, `tests/` (55 passed on the local database), `docs/data/`. |
| Backend (FastAPI, teammate) | BUILT | First version on its own branches: readings, validation, alert rules, doctor list. Still runs on SQLite; moving to the Supabase schema. |
| Alert rules and summaries | BUILT | In the backend (`api/app/analysis/`). Not yet writing alerts into Supabase. |
| ML model (next-day blood pressure) | BUILT | Model `mbp-v1`, trained on synthetic data. Test systolic error 5.2 mmHg vs 5.7 (7-day average) and 6.3 (yesterday). The prediction job is ready; running it on the cloud needs the database connection string. |
| ML phase 2: ring monitor, weekly doctor loop, weekly classifier | BUILT | Built as required by the backend review, on branch `AI`, 74 tests. Migration M7 is **live in the cloud** (approved 30 Oct); its tables fill once the jobs run there. See section 7.7. |
| Device: cuff, ring or bracelet | MISSING | No physical device chosen. Database tables for cuff readings and ring data are **live**; no device adapter built. |
| Real patient data | MISSING | None (on purpose). The demo patients are fictional, but their risk factors are drawn from real published Algerian statistics (STEPS 2016–2017). |
| Demo data in the cloud | LIVE | 14 accounts (1 doctor, 1 admin, 12 patients, 11 assigned), 730 readings, body measurements for everyone, ring data for 3 patients, 2 fasting days. All labelled `simulated`. Password in the local `.env` only. |

## 2. How everything connects




| Component | Talks to | Secret it uses | Where the secret lives |
|---|---|---|---|
| Frontend (patient and doctor) | FastAPI only (team decision A2) | none (the user's own login token) | nothing to store |
| FastAPI backend | Postgres; Supabase Auth for login; JWT check with JWKS (ES256) | `PULSE_DATABASE_URL`, `SUPABASE_URL`, anon key | server `.env` / hosting secrets |
| ML job, CSV import, seed | Postgres directly | `DATABASE_URL` or service-role key | data/AI machine or server `.env` (git-ignored) |
| Migrations | Supabase CLI `db push` | CLI login of the project owner | never shared |

**Never:** the service-role key or the database password in the frontend, in git, in chat or in screenshots. The GitHub repo is **public**, so anything committed is visible to everyone.

## 3. Who fills what

| Actor | What they enter or produce |
|---|---|
| Patient | Blood-pressure readings (manual), symptoms, meals / activity / sleep / stress, doses taken or missed, journal notes, trusted contacts, emergency-card details, consent choices, own devices. |
| Doctor | Care plan (target blood pressure, readings per day, instructions), prescribed medications, private notes and advice, alert review / resolve, journal review. |
| Device / import | Readings with device id and reading id, device quality flags, raw device payload; steps and sleep from a wearable. |
| System | Database trigger: validation status and flags, audit log. Rule engine (backend): alerts. ML job: predictions. Auth trigger: profile row on sign-up. |
| Admin | Roles (doctor / admin), doctor-patient assignments. Reads everything. |

## 4. Every table and field in the database

All 21 tables exist in the cloud project. Every table has the system fields `id` (except link tables) and `created_at`; these are filled automatically and not repeated below. **Blue** = patient, **purple** = doctor, **green** = system / device / admin.

## People and access

| Table | Purpose | Patient fills | Doctor fills | System / admin fills |
|---|---|---|---|---|
| **profiles** | One row per account | display_name, timezone (default Africa/Algiers), preferred_language (fr / ar) | display_name, timezone, preferred_language | id (= login id), role patient / doctor / admin (admin only), updated_at |
| **patient_profiles** | Medical background and risk factors | relevant_conditions (free list), condition_codes (controlled list), birth_year, sex, smoking_status, bread_portions_per_day, adds_salt_at_table, salty_processed_food, allergies, medication_context, regular_clinic, emergency_preferences, consent_location, data_consent_at | (reads) | user_id, regular_doctor_id (set by doctor/admin; read-only for patient), updated_at |
| **body_measurements** | Weight, height, waist (BMI, abdominal obesity) | measured_at, weight_kg, height_cm, waist_cm (source patient_reported) | same values (source clinician) | created_by; no edits or deletes |
| **doctor_patient_assignments** | Which doctor follows which patient. Drives all doctor access | (reads) | (reads) | doctor_id, patient_id, status active / ended, ended_at (admin / backend only) |
| **audit_log** | Who changed what | - | - | table_name, row_id, action, actor, changed_at, old_data, new_data (triggers; admin reads) |

## Readings and devices

| Table | Purpose | Patient fills | Device / import fills | System fills |
|---|---|---|---|---|
| **measurements** | Every blood-pressure reading (append-only) | systolic, diastolic, pulse, measured_at, measured_tz, body_position (sitting / standing / lying), arm (left / right), context (note), rested_5_min | device_id, ingestion_batch_id, external_id (device reading id), method, raw_payload, device quality flags | source (manual / device_sync / csv_import / simulated), provenance (measured / estimated / simulated), unit = mmHg, validation_status, quality_flags, validation_policy_version, created_by |
| **devices** | Cuffs, wearables, phone apps | device_type, manufacturer, model, external_device_id, label, bp_capability | - | patient_id (simulator devices only from seed) |

| | | | | |
|---|---|---|---|---|
| **wearable_samples** | Ring / bracelet / watch data | - | device_id, metric (heart_rate, hrv_rmssd, spo2, skin_temperature), value, unit, measured_at, external_id | source (device_sync / csv_import / simulated); unit and range checked per metric |
| **ingestion_batches** | One row per file import or device sync | - | source, device_id, file_name, file_sha256 (same file never imported twice) | rows_total / inserted / duplicate / rejected, rejections (row number + reason, no health values), created_by |

## Patient-reported context

| Table | Purpose | Patient fills | System fills |
|---|---|---|---|
| **symptom_reports** | Symptoms; checked independently of BP | symptoms: none, or any of chest_pain, shortness_of_breath, back_pain, weakness, numbness, vision_change, difficulty_speaking, severe_headache, confusion, dizziness, nausea, palpitations, other · onset_at · notes | reported_at, source (patient / bystander / clinician / simulated) |
| **lifestyle_logs** | Meals, activity, sleep, stress, fasting days | category, started_at, ended_at, description · meal: meal_type, sodium_mg + sodium_basis (label / estimated) · activity: activity_type, intensity, steps · sleep: sleep_quality 1-5 · stress: stress_level 1-5 | source (patient / device / simulated), updated_at |
| **medication_adherence_events** | Each dose taken / missed / skipped | medication_id, status, scheduled_for, taken_at (only if taken), note | source, reported_at |
| **journal_entries** | Notes / questions for the doctor | category, entry_text | doctor_review_status, reviewed_by, reviewed_at (filled by the doctor) |
| **trusted_contacts** | People to call in an emergency | name, relationship, phone, consented_at (contact agreed), active | - |
| **emergency_events** | Emergency screen history | trigger_reason, alert_id, symptoms_snapshot, started_at, ended_at, actions_attempted (logged as "tapped / opened", never "called"), reported_outcome, location_lat / lon (only if location_shared_with_consent) | - |

## Doctor-authored

| Table | Purpose | Doctor fills | Notes |
|---|---|---|---|
| **care_plans** | Targets and monitoring plan | target_systolic, target_diastolic, monitoring_readings_per_day, monitoring_instructions, lifestyle_recommendations, follow_up_instructions, clinician_guidance, effective_from / to, review_date, status, supersedes_id | One active plan per patient; a change = new row, old row becomes "superseded". Default target 140/90 when there is no plan. |

| | | | |
|---|---|---|---|
| **medications** | Prescriptions and patient-reported meds | name, dose_text, schedule_text (as written), active, started_on, ended_on (source = clinician, prescribed_by = doctor) | The patient may add own meds (source = patient_reported). PULSE never computes or changes doses. |
| **doctor_notes** | Private note + advice | note (private), advice, follow_up_at | Patients cannot read this table directly; the API shows them only advice and follow-up date. |

### Alerts and predictions (system)

| Table | Purpose | System fills | Patient / doctor actions |
|---|---|---|---|
| **alerts** | Rule results shown to patient and doctor | alert_class, rule_code, rule_version, reason, actions, persistent, measurement_id, symptom_report_id, trigger_source | Patient: acknowledge. Doctor: review, resolve, resolution_note. States: open → acknowledged → reviewed → resolved. |
| **alert_measurements** | Readings attached to one alert episode (e.g. the recheck) | alert_id, measurement_id, patient_id, attached_at | read only |
| **ml_predictions** | Model estimate of next-day morning BP | target_date, target, status (ok / insufficient_history), predicted_systolic / diastolic, intervals, model_version, trained_on, features_until, n_readings_used | Doctor reads only. Patients never see it. Never used for alerts. |

## 5. Ring, bracelet and cuff data

| Device | What it sends | How much to trust it | Where it goes in PULSE |
|---|---|---|---|
| **Upper-arm cuff** (validated, e.g. Omron with Bluetooth) | Systolic, diastolic, mean pressure, pulse, time, and status flags (body movement, cuff fit, irregular pulse, wrong position) through the standard Bluetooth Blood Pressure Profile (service 0x1810, measurement 0x2A35) | **Real measurement** if the model is clinically validated | `measurements`: source device_sync, provenance measured, method cuff_oscillometric; status flags → `quality_flags`; device → `devices` (bp_capability measured). fields LIVE adapter MISSING |
| **Smart ring / bracelet / watch** | Usually heart rate, heart-rate variability, SpO2, skin temperature, steps, sleep. Some show a **cuffless blood-pressure estimate** from the optical sensor (PPG), often calibrated with a cuff | **Estimate only.** ESH guidelines do not recommend cuffless devices for diagnosing or managing hypertension | BP estimate → `measurements` with provenance estimated, method wearable_estimate → automatically **needs_review** (flag estimated_not_cuff). Steps / activity → `lifestyle_logs` (activity, source device). Sleep → `lifestyle_logs` (sleep). Heart rate, HRV, SpO2, skin temperature → `wearable_samples` (unit and plausible range checked per metric) LIVE |

### How the data reaches PULSE

- **Cuff, direct Bluetooth:** phone browser with Web Bluetooth (Chrome / Edge on Android or desktop; not Safari / iPhone) → FastAPI → `measurements`.
- **Any device with a CSV export:** export file → our import tool (`python -m pulse_data.ingest import-csv`) BUILT. It checks columns and units, rejects bad rows with reasons and refuses the same file twice.
- **Ring / bracelet vendor cloud or SDK:** depends on the brand. Many need a developer account or partnership. Choose the device first, then check its export options.

### Rules for ring / bracelet data

- Never show a ring estimate as a measured blood pressure. Always label it "estimate".
- Estimates are `needs_review`, so they are left out of averages, trends and ML training (team rule: valid readings only).
- A high estimate should ask the patient for a **cuff reading**. It should not trigger the emergency path on its own; symptoms always can. clinician to confirm
- Heart rate / HRV / SpO2 / skin temperature go to `wearable_samples` (patient, device, metric, value, unit, time, source). Written by the backend only; demo data has a simulated ring for Karim, Nadia and Fatima. LIVE

**Recommendation for the demo:** a validated upper-arm cuff gives real numbers that the database already handles. A ring is good for activity and sleep context, not for the blood-pressure value itself.

## 6. Do the fields match the Algerian factors that affect blood pressure?

**What the Algerian national survey shows** (WHO STEPwise 2016–2017, analysis by Moussouni et al., *BMC Public Health* 2022; 6,765 adults aged 18–69):

- Hypertension **31.6%**, prehypertension **36.2%**. Hypertension is higher in women (34.2%) than men (28.3%).
- **41.1%** of people with hypertension did not know they had it.
- Independent risk factors for hypertension: **age** (60–69 vs 18–29: odds ratio 12.3), **hypercholesterolemia** (2.0), **obesity** BMI ≥ 30 (1.7), **abdominal obesity** (1.7), separated / divorced (2.9).
- Common but not significant after adjustment: daily smoking 11.1%, physical inactivity 19.4%, diabetes / high blood sugar 17.5%. Obesity 24.9%, overweight 35.0%, abdominal obesity 67.8%.

**Salt:** bread is described as a major salt source in western Algeria (low-salt bread study). A pilot study in Casablanca bakeries (Morocco) found about 17 g of salt per kg of bread, roughly 8–9 g of salt a day from bread alone. We found no measured national salt-intake figure for Algeria.
**Ramadan:** small studies show blood pressure mostly unchanged when treatment continues. Medication timing (iftar or suhoor) varies, and fasting brings a risk of dehydration. Any change of timing or dose is the doctor's decision only.

| Factor | Why it matters | Status | Where it lives (migrations M5 / M6, live in the cloud) |
|---|---|---|---|
| Age | Strongest factor | YES | `patient_profiles.birth_year` (year only, data minimisation) |
| Sex | Hypertension higher in women | YES | `patient_profiles.sex` female / male |
| Weight, BMI, waist | Obesity and abdominal obesity are significant | YES | table `body_measurements`: weight_kg, height_cm, waist_cm (BMI computed; abdominal obesity = waist > 94 cm men / > 80 cm women) |
| Cholesterol, diabetes, kidney / heart disease | Cholesterol significant; diabetes common | YES | `patient_profiles.condition_codes`: hypertension, diabetes, dyslipidemia, chronic_kidney_disease, heart_disease, stroke_history, pregnancy |
| Salt / bread | Main dietary salt source | YES | `bread_portions_per_day`, `adds_salt_at_table`, `salty_processed_food` (never … always, STEPS style) + meal `sodium_mg` when known |
| Smoking | 11.1% smoke daily; cardiovascular risk | YES | `patient_profiles.smoking_status` never / former / current |
| Physical activity | 19.4% inactive | YES | lifestyle_logs: activity_type, intensity, steps |
| Sleep, stress | Context for the doctor | YES | lifestyle_logs: sleep_quality, stress_level |
| Medication + adherence | Key for control | YES | medications + adherence events (Algerian brand names as free text) |
| Ramadan fasting | Meal and medication timing change | YES | lifestyle_logs category `fasting` (start of fast → iftar); medication timing stays the doctor's decision |

| | | | |
|---|---|---|---|
| Local time (morning / evening) | Morning readings, daily windows | YES | timezone default Africa/Algiers |
| Language | French / Arabic wording | YES | `profiles.preferred_language` fr / ar |
| Emergency number | Shown on the emergency screen | TO VERIFY | backend config (14 or 1021); verify with a local source; never dial in tests |
| Undiagnosed patients (41.1%) | Screening need | out of scope | PULSE follows known patients; mention in the pitch as future work |

**Demo data:** 8 of the 12 demo patients have their sex, age band, smoking, diabetes, cholesterol, BMI band and abdominal obesity drawn from the published STEPS shares above. Height, salt-habit answers, language and ring values are labelled assumptions ( `docs/data/DATA_SOURCES.md` ). The ML MVP still uses blood-pressure history only (team contract); the new fields serve the doctor view first.

# 7. The full logic

## 7.1 Life of one reading

| 1. Enter | 2. Format check | 3. Validate | 4. Rules | 5. Alert | 6. Show | 7. Doctor | 8. ML |
|---|---|---|---|---|---|---|---|
| Patient types it (or device / CSV) | FastAPI / importer: numbers, unit mmHg, time | DB trigger pulse-v1 sets valid / needs_review / invalid + flags | Backend rules evaluate reading + symptoms | Create or attach to an open alert (no duplicates) | Next step for the patient; emergency screen if red flags | Patient list sorted: emergency > severe > trend > notice | Batch job: next-day estimate for the doctor |

## 7.2 Validation policy `pulse-v1` (database trigger, applies to every reading)

| Result | Rule |
|---|---|
| **invalid** (stored, not used) | systolic ≤ diastolic · systolic outside 50–300 · diastolic outside 25–200 · pulse outside 25–250 · more than 5 minutes in the future |
| **needs_review** (stored, shown with flags) | systolic outside 70–250 or diastolic outside 40–150 · estimated (not a cuff) · same values within 2 minutes (double entry) · device reported a problem |
| **valid** | everything else. A very high reading (e.g. 186/118) is **valid data**; it is an alert matter, not a data problem. |

Open point: the backend's own Python validation uses diastolic ≥ 30 and a pulse-pressure check (20–120). The team contract makes the database trigger the single source of truth. Whether to add a pulse-pressure flag to the trigger is a clinician question.

## 7.3 Alert rules (backend `rules.py`, deterministic, independent of ML)

| Rule | When | Level | What happens |
|---|---|---|---|
| R-EMG-1 | Any red-flag symptom: chest_pain, shortness_of_breath, back_pain, numbness, weakness, vision_change, difficulty_speaking | emergency | Emergency screen immediately, whatever the reading. Raises a recent severe alert instead of making a second one. |
| R-SEV-1 | systolic ≥ 180 or diastolic ≥ 120 | severe | Symptom check, then recheck after 60 seconds |
| R-SEV-2 | Recheck within 15 min still ≥ 180 / ≥ 120 | severe, persistent | Attached to the same alert; urgent guidance |
| R-SEV-3 | Recheck within 15 min below 180 / 120 | severe | Alert stays open for the doctor; reading attached |

| | | | |
|---|---|---|---|
| R-TGT-1 | Reading ≥ target (care plan, default 140/90) | notice | One active notice per patient |
| R-TRD-1 | ≥ 3 valid readings in 7 days and their mean ≥ target | trend | One active trend alert per patient |

Alert states: open → acknowledged (patient or doctor) → reviewed (doctor) → resolved (doctor; a note is required for severe / emergency). Expected mapping to the database `alert_class`: emergency → possible_emergency, severe → severely_high, notice → above_target, trend → above_target_trend **confirm with backend**.

## 7.4 Summary numbers (backend `summary.py`)

- 7-day and 30-day average systolic / diastolic, 7-day maximum, % of readings above target, days with a reading. **Valid readings only.** needs_review and invalid readings are counted, not averaged.
- Trend: last 7 days vs the 7 days before, at least 3 readings in each. Systolic change ≥ +5 → rising; ≤ −5 → falling; else stable; otherwise "insufficient data".

## 7.5 ML model (next task, data/AI)

| Item | Decision (team contract §4) |
|---|---|
| Predicts | Next day's **morning** mean systolic and diastolic (readings before 12:00 local time) |
| Inputs | Valid readings from before the target day only; blood-pressure history + day of week (no lifestyle in the MVP) |
| Eligibility | ≥ 6 valid readings on ≥ 4 days in the last 7 days and ≥ 1 morning reading; otherwise status "insufficient_history" |
| Training data | Synthetic set `pulse-sim-v1` (200 patients × 90 days). Labelled synthetic everywhere |
| Testing | Patients split 70 / 15 / 15 (no patient in two sets), time order kept. Compared with two baselines: yesterday's morning value and the 7-day mean |
| Models | Regularised linear regression and gradient boosting; keep the best on validation, test once |
| Metrics | MAE and RMSE in mmHg with 95% confidence intervals over patients |
| Output | `ml_predictions`, doctor-only panel "Model estimate (synthetic training data)" |
| Never | Writes alerts or measurements, decides an emergency, or suggests medication changes |

## 7.7 ML phase 2: live ring, weekly doctor loop, classification (built 30 Oct, local)

**1. Ring**
Samples arrive in `wearable_samples` (simulated stream for now)

**2. Monitor**
Live 10-minute and night windows; compared with medical references, the personal normal (28 nights) or the doctor's band

**3. Flags**
`wearable_flags`: advisory, never alerts, never the safety path

**4. Weekly**
Report per patient: home-BP category, control vs target, P(next week above target), ring summary, proposed bands

**5. Doctor**
Approves or edits bands (`approve_band`), confirms or dismisses flags, reviews the report

**6. Learn**
Doctor labels retune personal thresholds only; recall gate; no automatic deployment

| Model | Result on synthetic test patients | Status |
|---|---|---|
| `ring-v1` ring monitor | Personal baseline catches 10 / 11 injected episodes (0.22 false flags per patient-week); medical ranges alone catch 3 / 11 | advisory, doctor only |
| `wcls-v1` weekly classifier | AUROC 0.977 vs 0.907 for "status persists", better calibrated; but no skill on weeks where the status changes, so it is shown as "how close to the line" | in the weekly report |
| `mbp-v2` (ring features) | 5.20 vs 5.23 mmHg for v1: no real gain | not served; v1 stays |

Safety: the ring rule R-RING-1 (SpO2 below 94 % over 10 minutes → prompt a cuff reading + symptom check) is specified for the **backend** to run directly on raw samples, so the safety path never depends on an ML job. Home-BP categories use the ACC/AHA 2017 **home** values (135/85 = office 140/90), never office grades.

### 7.6 Safety rules that hold everywhere

- No diagnosis, no prescription, no advice to start / stop / change medication.
- Symptoms trigger the emergency path on their own; ML never delays urgent guidance.
- Nothing is silently deleted: wrong readings stay, flagged. Readings cannot be edited by app users.
- Synthetic data is always labelled; ring estimates are always labelled.
- Health values never go into logs or error messages.

## 8. What we need

| Item | Why | Who | Status |
|---|---|---|---|
| Turn off public sign-up | Otherwise anyone with the public key can create accounts | Project owner | DONE verified: sign-up refused |
| Approve the backend architecture and contract (D-1 to D-4), merge PR #1 | Everyone builds on the same base | Backend lead | OPEN |
| Demo data in the cloud (password from `PULSE_DEMO_PASSWORD`, never committed) | Shared data for frontend and demo | Data/AI | DONE 14 accounts, 730 readings |
| Share the demo password with the team privately (it is in the local `.env`) | Teammates log in as demo users | You | OPEN |
| Cloud ML predictions: put the cloud `DATABASE_URL` in the local `.env`, then run the prediction job | Doctor prediction panel | You (password) + data/AI | OPEN |
| Clinician review: 180/120, red-flag list, validation limits, wording | Safe, honest alerts | Medical mentor | OPEN |
| Verified Algerian emergency number + demo substitute number | Emergency screen | Team + mentor | TO VERIFY |
| Device decision (validated cuff recommended) | Real readings in the demo | Team | OPEN |
| Algerian fields (section 6) + ring table (section 5) | Age, BMI, smoking, conditions, salt, fasting, language, wearables | Data/AI | DONE M5 + M6 live; backend to review |
| Hosting for FastAPI (HTTPS) | Demo URL, location permission | Backend lead | OPEN |
| ML pipeline (training, evaluation, prediction job, model card) | Doctor prediction panel | Data/AI | DONE `ml/` |
| Real consented readings (later) | Real model accuracy | Team | MISSING no open dataset fits |
| Security scan key (HAWK_API_KEY), branch protection on main | Optional hardening | Repo owner | OPEN |

## 9. Next steps, in order

| # | Step | Who | Depends on |
|---|---|---|---|
| 1 | Tell the backend that M5 + M6 and the demo data are live (`docs/data/MIGRATIONS.md`); share the demo password privately | You | - |
| 2 | Backend maps the new fields and tables in its API (profile, body measurements, wearable read); replays alerts on the demo history | Backend | 1 |
| 3 | Fill `ml_predictions` in the cloud: cloud `DATABASE_URL` in the local `.env`, then `python -m pulse_ml.predict --all` | You + data/AI | - |

| | | | |
|---|---|---|---|
| 4 | Choose a real device (validated cuff recommended); build its adapter | Team, then data/AI | - |
| 5 | Screens for the new fields: profile (age, sex, smoking, conditions, salt habits), weight / waist entry, fasting days, ring data chart | Frontend | 2 |
| 6 | Frontend screens on the API contract (reading, symptoms, emergency, doctor list, prediction panel) | Frontend | 3 |
| 7 | Clinician review and emergency number check | Team + mentor | - |
| 8 | Full demo from a fresh reset: seed → alerts → predictions → demo script | Everyone | 2–7 |

## Sources


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7. European Society of Hypertension: cuffless BP devices statement (J Hypertens 2022;40(8):1449-1460) and validation recommendations (J Hypertens 2023;41(12):2074-2087). https://pubmed.ncbi.nlm.nih.gov/37303198/
8. PULSE team documents: `PULSE_INTEGRATION_CONTRACTS.md`, `PULSE_BLOCKERS_AND_REQUIREMENTS.md` (branch docs/backend-architecture); `docs/data/` (branch AI); backend `api/app/analysis/rules.py`, `summary.py`.


Study figures are quoted from the sources above. Thresholds and proposals in this guide are prototype choices awaiting clinician review.